

Comparitive Performance Report for tentacle-3-way-replica-balance-reads-off vs direct-reads-pr2-3-way-replica-balance- reads-off-rerun2 vs direct-reads-pr2-3-way-replica-balance- reads-on-split-off-rerun2 vs direct-reads-pr2-3-way-replica-balance- reads-on-split-256k-rerun2

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Comparison summary for
tentacle-3-way-replica-balance-reads-off vs
direct-reads-pr2-3-way-replica-balance-reads-off-rerun2 vs
direct-reads-pr2-3-way-replica-balance-reads-on-split-off-rerun2 vs
direct-reads-pr2-3-way-replica-balance-reads-on-split-256k-rerun2

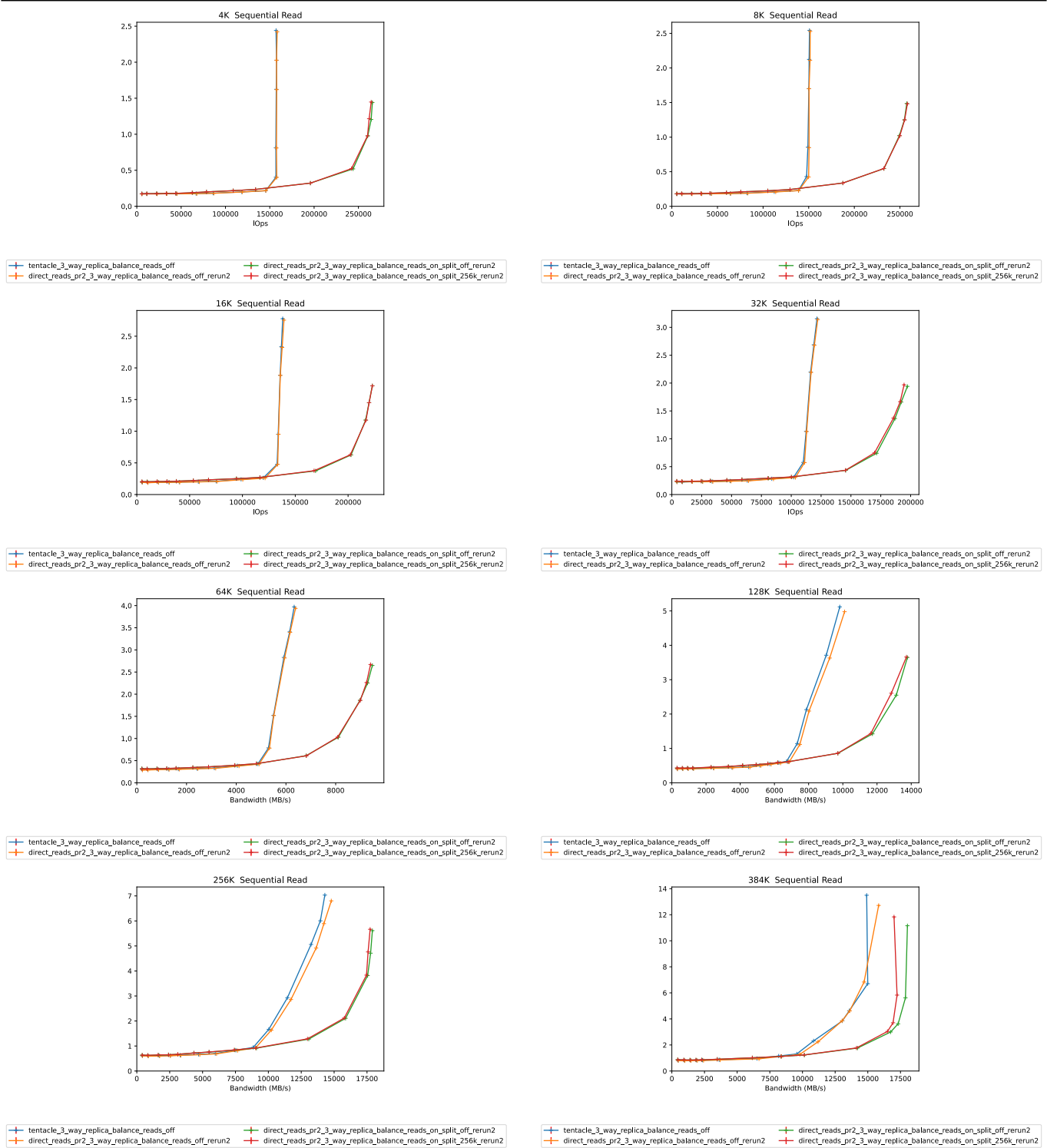
Sequential							
Read	tentacle_3_way-replica-balance-reads-off	direct-reads-pr2-3-way-replica-balance-reads-off-rerun2	direct-reads-pr2-3-way-replica-balance-reads-on-split-off-rerun2	direct-reads-pr2-3-way-replica-balance-reads-on-split-256k-rerun2	Change	Change	Change
4K	157548@2.0ms	158346@2.4	1%	265717@1.4	69%	264393@1.4	68%
8K	150852@2.5ms	151639@2.5	1%	257455@1.5	71%	258173@1.5	71%
16K	138108@2.8ms	139269@2.8	1%	222946@1.7	61%	222987@1.7	61%
32K	121533@3.2ms	122178@3.1	1%	197285@1.9	62%	194498@2.0	60%
64K	6331@4.0ms	6385@3.9	1%	9476@2.6	50%	9403@2.7	49%
128K	9823@5.1ms	10103@5.0	3%	13777@3.6	40%	13721@3.7	40%
256K	14295@7.0ms	14785@6.8	3%	17908@5.6	25%	17736@5.7	24%
384K	14995@6.7ms	15830@12.7	6%	18034@11.2	20%	17239@5.8	15%
512K	17061@15.7ms	16707@8.0	-2%	18207@7.4	7%	14796@9.1	-13%
768K	17212@23.4ms	15548@8.1	-10%	17904@22.5	4%	15673@8.0	-9%
1024K	16689@24.1ms	15688@8.5	-6%	17015@15.8	2%	15290@4.4	-8%
2048K	14689@9.1ms	13053@10.3	-11%	14666@9.1	-0%	16067@8.3	9%
4096K	13597@59.2ms	12305@10.9	-10%	14023@57.4	3%	14568@9.2	7%

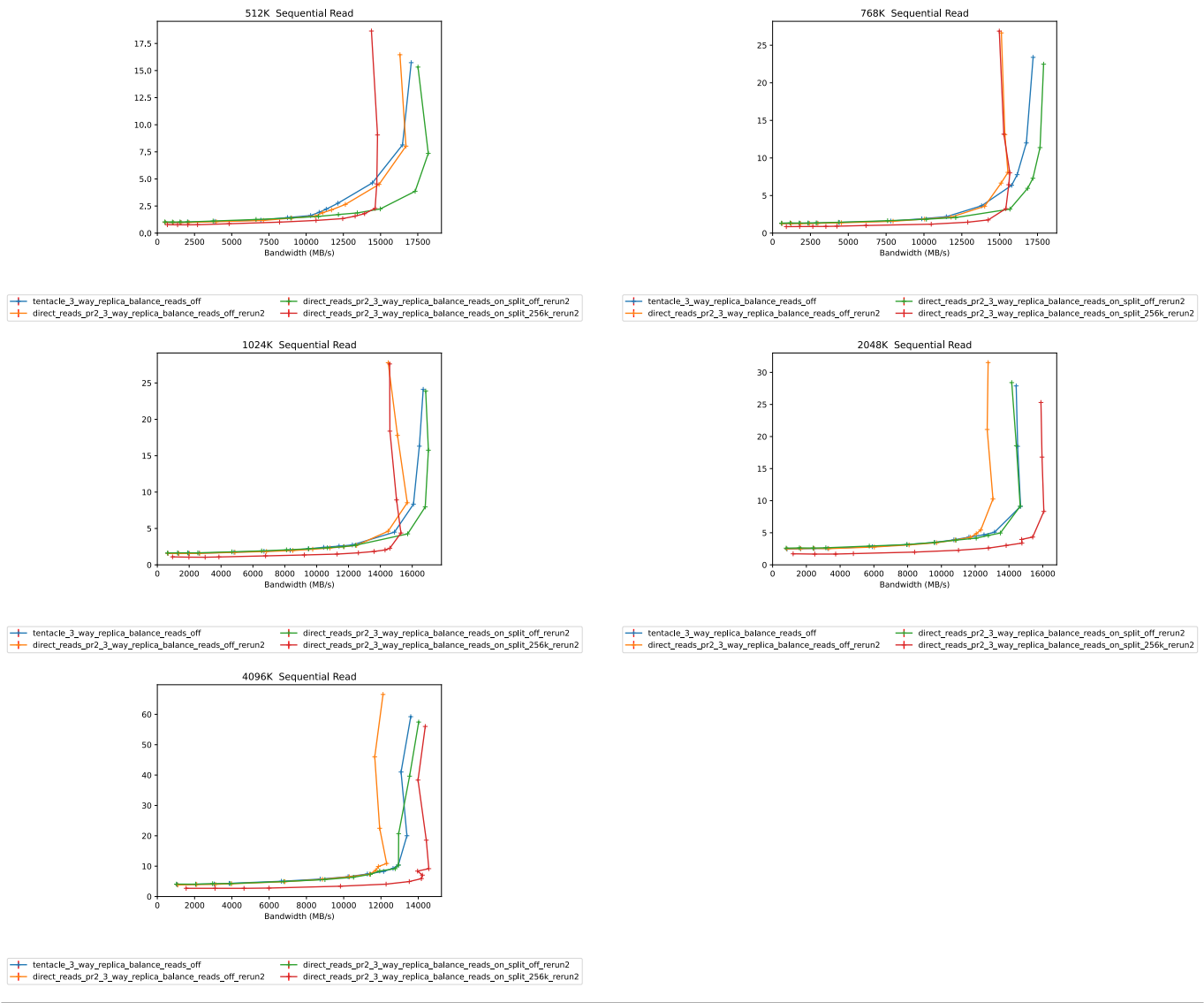
Random							
Read	tentacle_3_way-replica-balance-reads-off	direct-reads-pr2-3-way-replica-balance-reads-off-rerun2	direct-reads-pr2-3-way-replica-balance-reads-on-split-off-rerun2	direct-reads-pr2-3-way-replica-balance-reads-on-split-256k-rerun2	Change	Change	Change
4K	219491@1.2ms	233097@1.6	6%	250297@1.0	14%	250356@1.3	14%
8K	209044@1.2ms	228795@1.7	9%	235241@1.1	13%	230551@1.1	10%
16K	188007@1.4ms	203900@1.9	8%	211260@1.8	12%	208560@1.5	11%
32K	167762@2.3ms	181336@1.8	8%	186854@1.7	11%	189493@1.3	13%
64K	9032@2.8ms	9603@1.7	6%	9705@2.2	7%	9699@2.6	7%
128K	13244@3.8ms	14375@3.5	9%	14259@2.3	8%	13922@2.4	5%
256K	16481@4.1ms	18121@3.7	10%	17361@4.8	5%	17190@5.8	4%
384K	17128@11.7ms	17080@2.9	-0%	18636@5.4	9%	17698@11.4	3%
512K	17600@15.2ms	17128@3.9	-3%	17472@3.8	-1%	15681@17.1	-11%
768K	17110@23.5ms	16397@3.1	-4%	17061@5.9	-0%	15858@3.2	-7%
1024K	16507@16.2ms	14922@9.0	-10%	17273@15.5	5%	15949@25.2	-3%
2048K	14724@18.2ms	13509@9.9	-8%	14964@26.9	2%	16031@8.3	9%
4096K	13969@57.6ms	12830@10.4	-8%	14102@57.1	1%	14976@17.9	7%

Comparison summary for tentacle-3-way-replica-balance-reads-off vs							
direct-reads-pr2-3-way-replica-balance-reads-off-rerun2 vs							
direct-reads-pr2-3-way-replica-balance-reads-on-split-off-rerun2 vs							
direct-reads-pr2-3-way-replica-balance-reads-on-split-256k-rerun2							
Read/Write	tentacle_3_way-direct-reads-pr2-3-way-replica-balance-reads-off-rerun2	direct-reads-pr2-3-way-replica-balance-reads-off-rerun2	direct-reads-pr2-3-way-replica-balance-reads-on-split-off-rerun2	direct-reads-pr2-3-way-replica-balance-reads-on-split-256k-rerun2	direct-reads-pr2-3-way-replica-balance-reads-off-rerun2	direct-reads-pr2-3-way-replica-balance-reads-on-split-off-rerun2	direct-reads-pr2-3-way-replica-balance-reads-on-split-256k-rerun2
4K_70/30	119218@3.2ms	121693@3.1	2%	130014@2.9	9%	129777@3.0	9%
16K_70/30	101882@3.8ms	105266@3.6	3%	106584@3.0	5%	108980@3.5	7%
64K_30/70	3170@5.3ms	3184@5.2	0%	2995@5.6	-6%	3182@7.9	0%
64K_70/30	4847@4.3ms	5084@4.9	5%	5056@4.1	4%	5112@4.9	5%

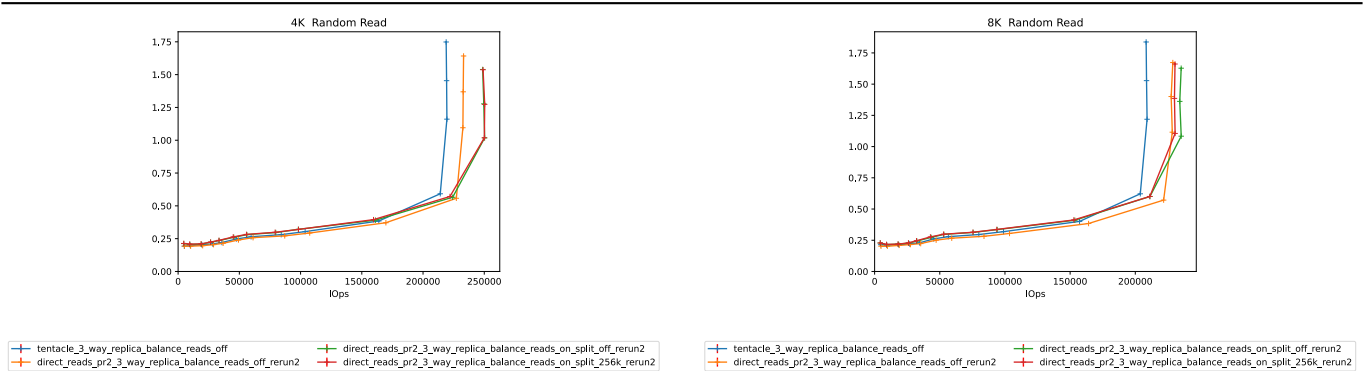
Response Curves

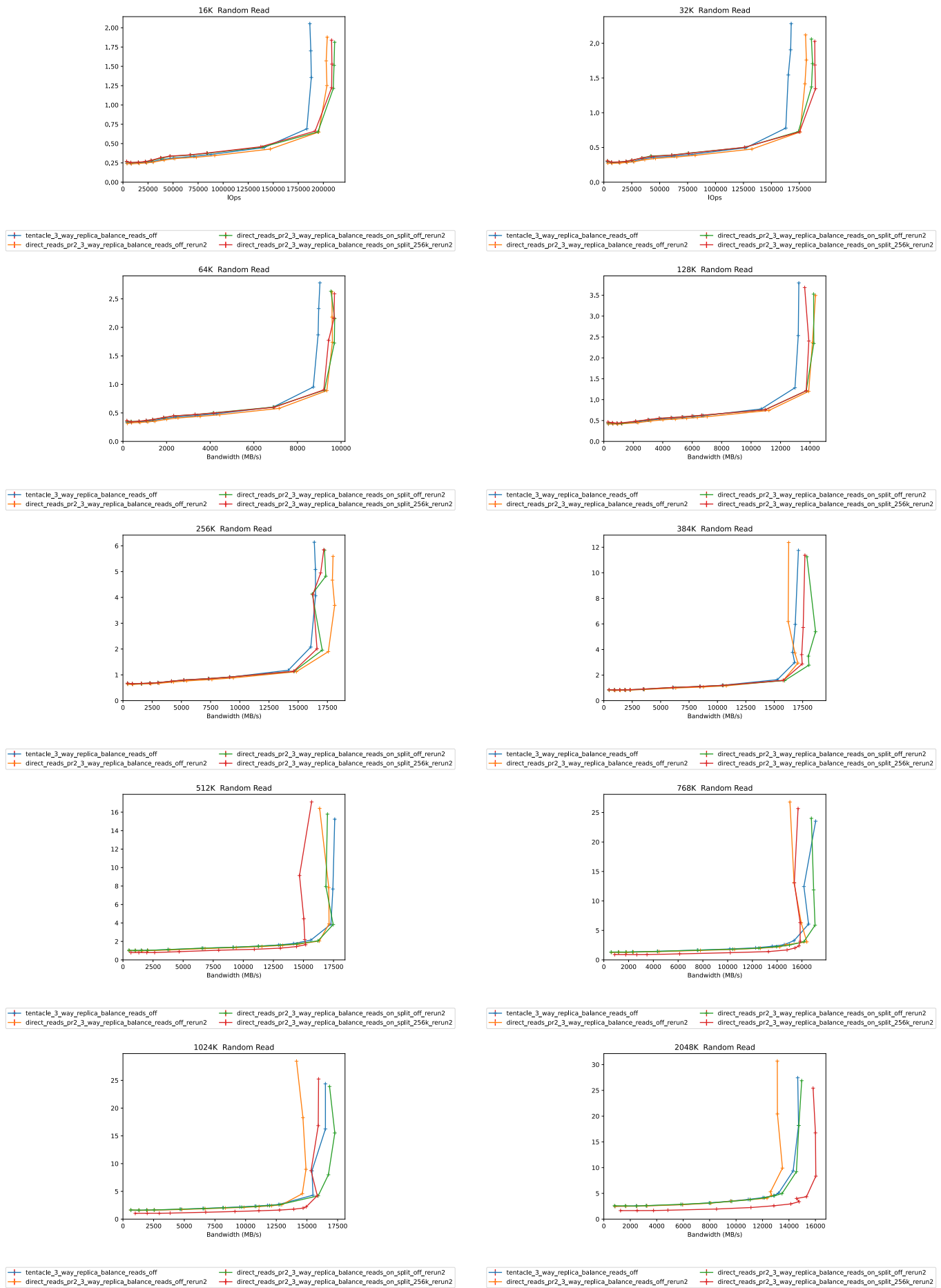
Sequential Read





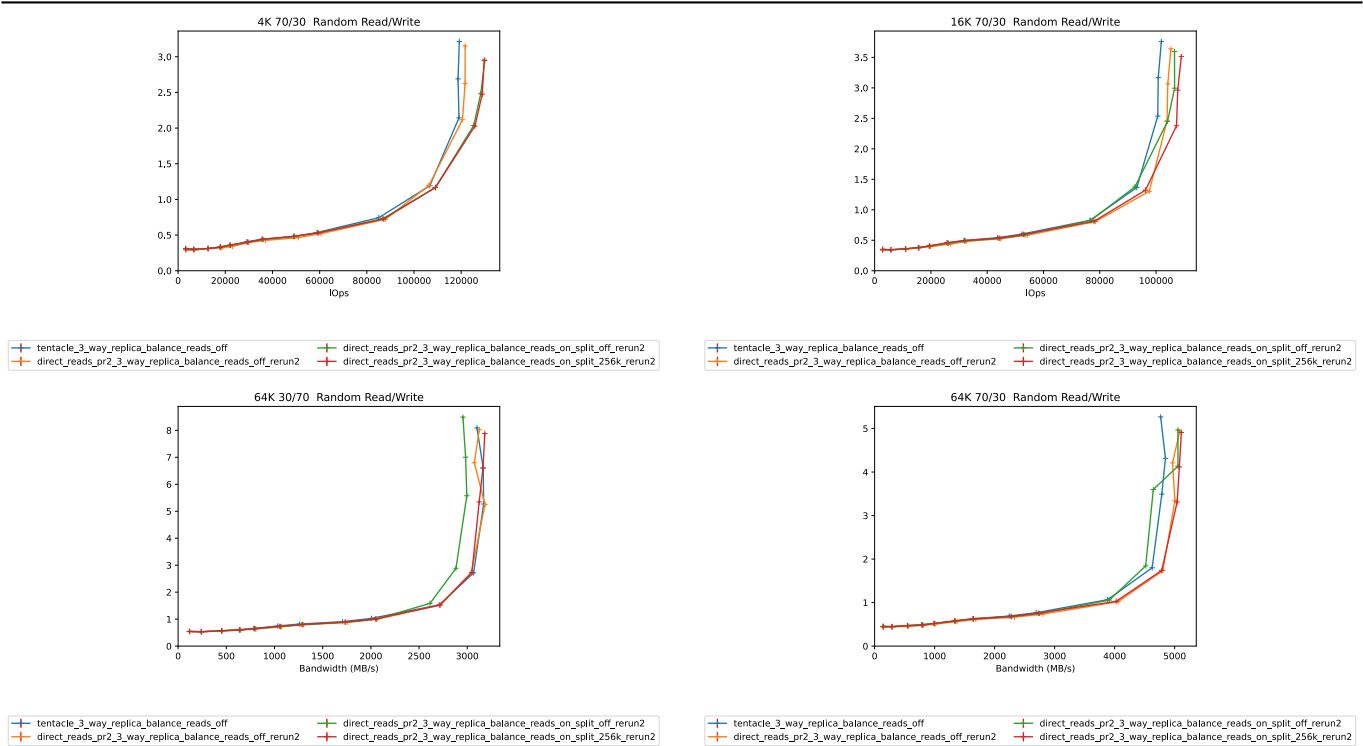
Random Read







Random Read/Write



Configuration yaml files

Only yaml files that differ by more than 20 lines from the yaml file for the baseline directory will be added here in addition to the baseline yaml

results

```

librbd fio:
  cmd_path: /usr/local/bin/fio2
  create_report: true
  fio_out_format: json
  log_avg_msec: 100
  log_bw: true
  log_iops: true
  log_lat: true
  norandommap: true
  osd_ra:
    - 4096
  poolname: rbd_replicated
  prefill:
    blocksize: 64k
    numjobs: 1
  procs_per_volume:
    - 1
  ramp: 30
  rbdname: cbt-librbd fio
  time: 90
  time_based: true
  use_existing_volumes: true
  vol_size: 1000
  volumes_per_client:
    - 16
  wait_pgautoscaler_timeout: 10
  workloads:
    128k randomread:
      jobname: randread
      mode: randread
      numjobs:
        - 1
      op_size: 131072
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8
        - 12
        - 16
        - 20
        - 24
        - 28
        - 32
        - 64
        - 128
        - 256
        - 384
    128k sequentialread:
      jobname: seqread
      mode: read
      numjobs:
        - 1
      op_size: 131072
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8

```

```

- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 256
- 384
16kmixread70:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 16384
  rwmixread: 70
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
16krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 16384
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
16ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 16384
  total_iodepth:
    - 1
    - 2
    - 4

```

```

- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
1Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 1048576
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 256
    - 384
1Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 1048576
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 256
    - 384
256krandomread:
  jobname: randomread
  mode: randread
  numjobs:
    - 1
  op_size: 262144
  total_iodepth:
    - 1

```

```

- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
256ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 262144
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
2Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 2097152
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 192
2Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 2097152
  total_iodepth:
    - 1

```

- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192

32krandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 32768
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

32ksequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 32768
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

384krandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 393216
- total_iodepth:
- 1

- 2
- 3
- 4
- 5
- 8
- 16
- 24
- 32
- 64
- 128
- 160
- 256
- 512

384ksequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 393216
- total_iodepth:
- 1
- 2
- 3
- 4
- 5
- 8
- 16
- 24
- 32
- 64
- 128
- 160
- 256
- 512

4Mrandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 4194304
- total_iodepth:
- 1
- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192

4Msequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 4194304
- total_iodepth:
- 1

- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192

4kmixread70:

- jobname: mixread
- mode: randrw
- numjobs:
- 1
- op_size: 4096
- rwmixread: 70
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

4krandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 4096
- total_iodepth:
- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

4ksequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 4096
- total_iodepth:

- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384

512krandomread:

- jobname: randread
- mode: randread
- numjobs:
- 1
- op_size: 524288
- total_iodepth:
- 1
- 2
- 3
- 4
- 8
- 16
- 24
- 32
- 40
- 48
- 64
- 128
- 256
- 512

512ksequentialread:

- jobname: seqread
- mode: read
- numjobs:
- 1
- op_size: 524288
- total_iodepth:
- 1
- 2
- 3
- 4
- 8
- 16
- 24
- 32
- 40
- 48
- 64
- 128
- 256
- 512

64kmixread30:

- jobname: mixread
- mode: randrw
- numjobs:
- 1
- op_size: 65536
- rwmixread: 30

```

    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  64kmixread70:
    jobname: mixread
    mode: randrw
    numjobs:
      - 1
    op_size: 65536
    rwmixread: 70
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  64krandomread:
    jobname: randread
    mode: randread
    numjobs:
      - 1
    op_size: 65536
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  64ksequentialread:
    jobname: seqread
    mode: read
    numjobs:
      - 1

```

```

  op_size: 65536
  total_iodepth:
  - 1
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
  - 24
  - 32
  - 64
  - 128
  - 256
  - 320
  - 384
768krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 786432
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 24
  - 32
  - 40
  - 48
  - 64
  - 128
  - 256
  - 512
768ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
  - 1
  op_size: 786432
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 5
  - 8
  - 16
  - 24
  - 32
  - 64
  - 128
  - 160
  - 256
  - 512
8krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1

```

```

    op_size: 8192
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  8ksequentialread:
    jobname: seqread
    mode: read
    numjobs:
      - 1
    op_size: 8192
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  precondition:
    jobname: precond1rw
    mode: randwrite
    monitor: false
    numjobs:
      - 1
    op_size: 65536
    time: 600
    total_iodepth:
      - 16
cluster:
  archive_dir: /tmp/cbt
  ceph_mgr_cmd: /usr/bin/ceph-mgr
  ceph_mon_cmd: /usr/bin/ceph-mon
  ceph_osd_cmd: /usr/bin/ceph-osd
  ceph_run_cmd: /usr/bin/ceph-run
  ceph_cmd: /usr/bin/ceph
  clients:
    - --- server1 ---
  clusterid: ceph
  conf_file: /cbt/ceph.conf.4x1x1.fs
  fs: xfs
  head: --- server1 ---
  iterations: 1
  mgrs:
    --- server1 ---:

```

```
    a: null
mkfs_opts: -f -i size=2048
mons:
  --- server1 ---:
    a: --- IP Address ---:6789
mount_opts: -o inode64,noatime,logbsize=256k
osds:
- --- server1 ---
  osds_per_node: 6
  pdsh_ssh_args: -a -x -l%u %h
  rados_cmd: /usr/bin/rados
  rbd_cmd: /usr/bin/rbd
  tmp_dir: /tmp/cbt
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -F0 --rawtoo --sep ";" -f {collectl_dir}
```